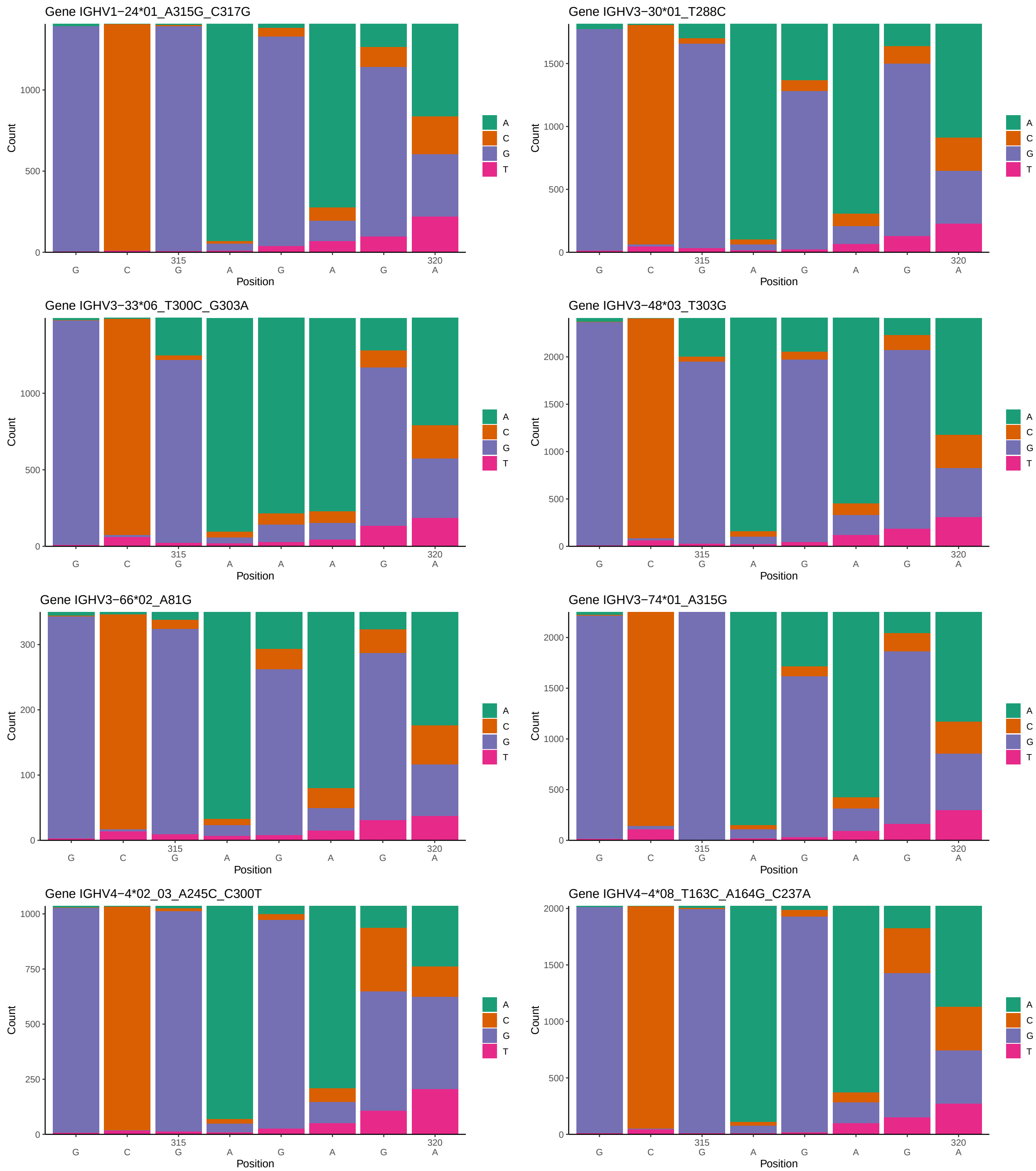
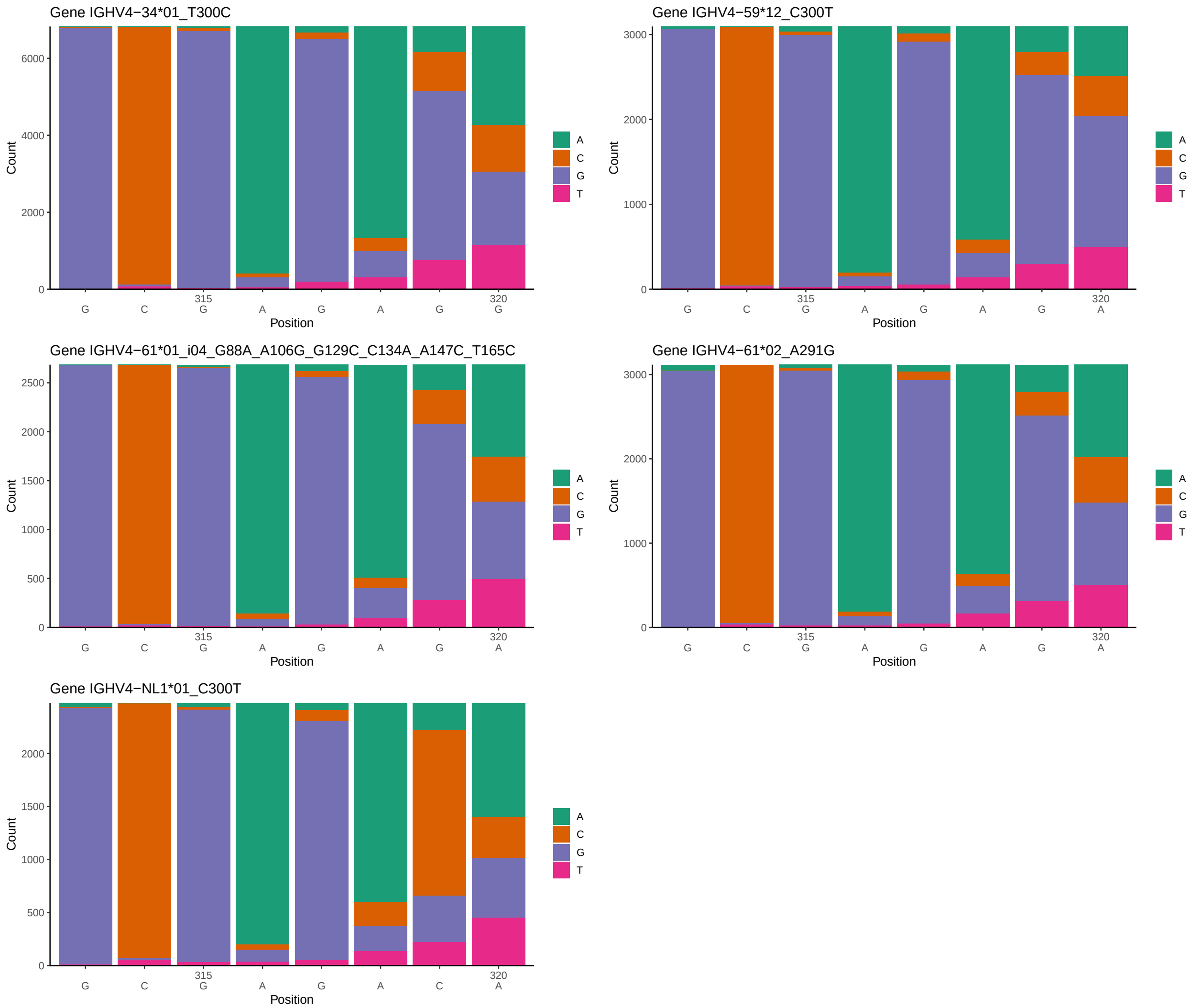
OGRDBstats Report

Contents

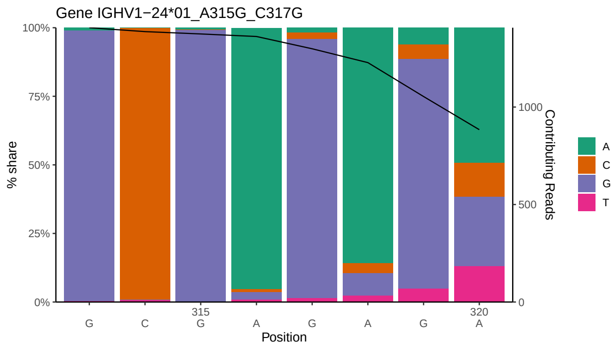
1 Novel sequence analysis

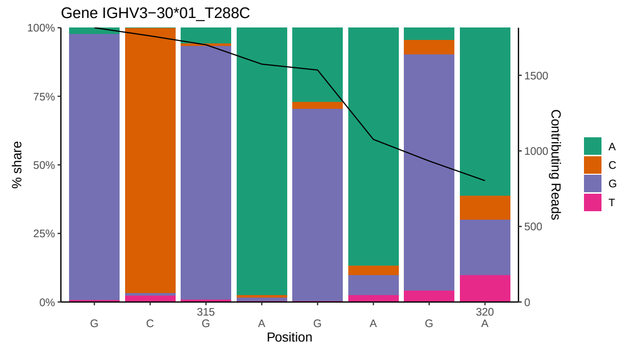
1.1 End-nucleotide composition

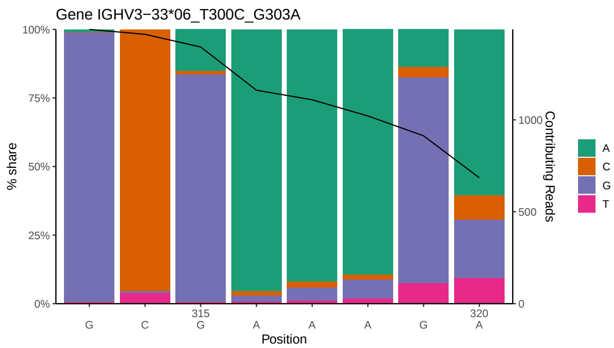


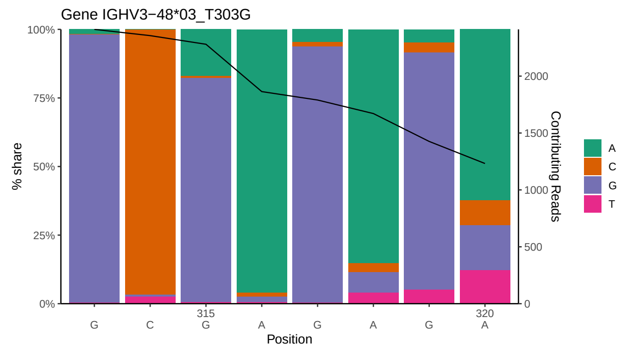

Gene IGHV4−34*01_T300C
Gene IGHV4−59*12_C300T
Gene IGHV4−61*01_i04_G88A_A106G_G129C_C134A_A147C_T165C
Gene IGHV4−61*02_A291G
Gene IGHV4−NL1*01_C300T
Count
Position
A
C
G
T
315
320

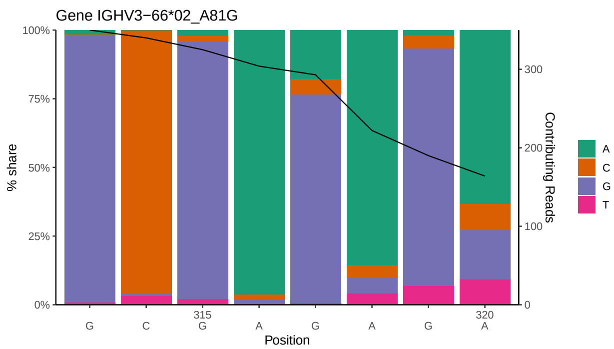
1.2 Per-nucleotide consensus where previous nucleotides match the consensus

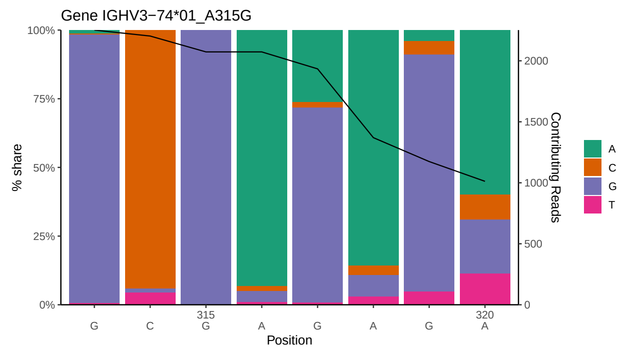


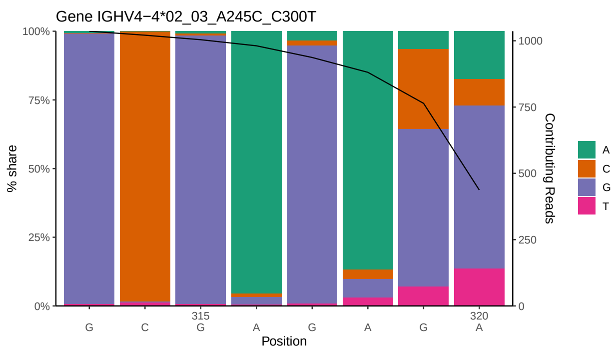


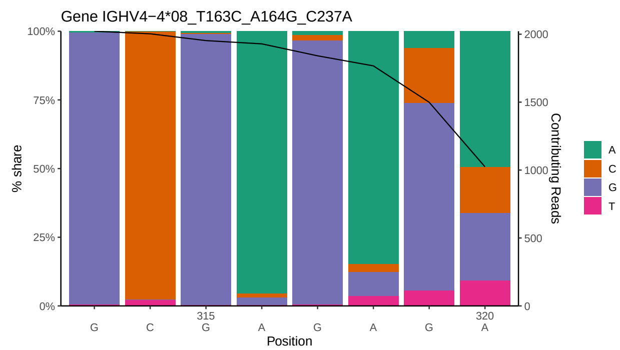


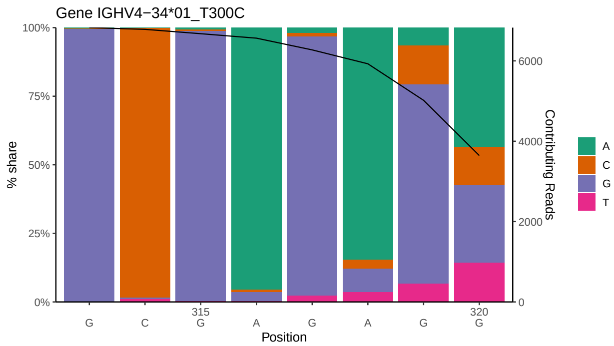


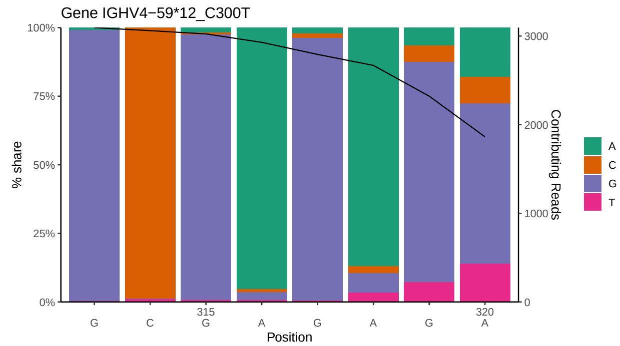


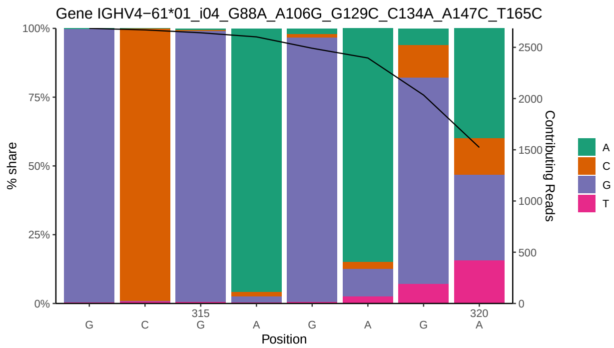


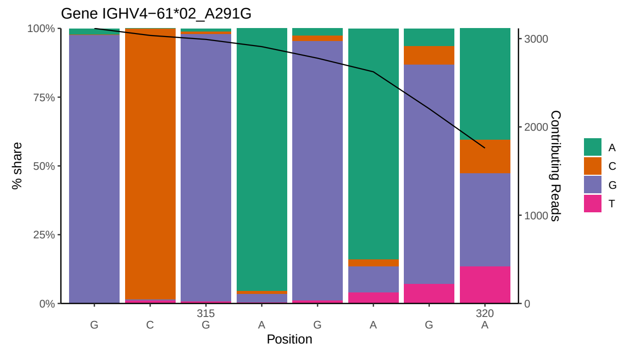


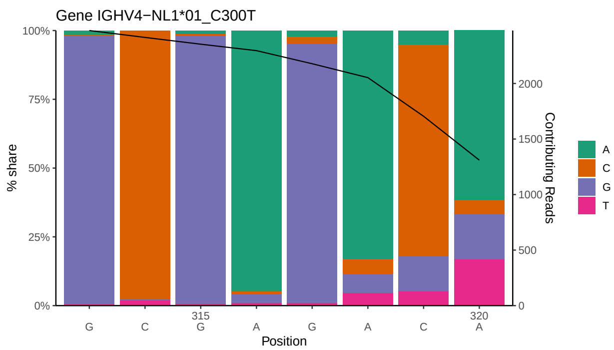



Gene IGHV4−34*01_T300C
% share
100%
75%
50%
25%
0%
Contributing Reads
6000
4000
2000
0
G C G A G A G G
315
320
Position
A
C
G
T

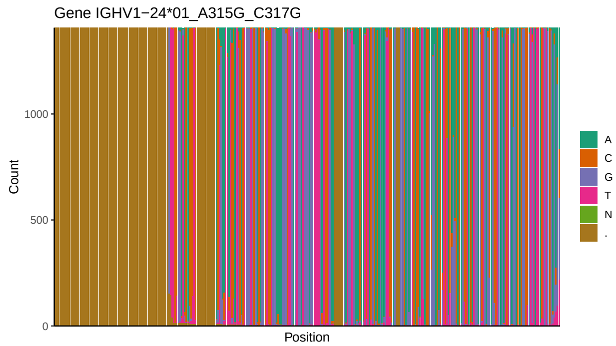

Gene IGHV4−59*12_C300T
% share
100%
75%
50%
25%
0%
Contributing Reads
3000
2000
1000
0
G C G A G A G A
315
320
Position
A
C
G
T

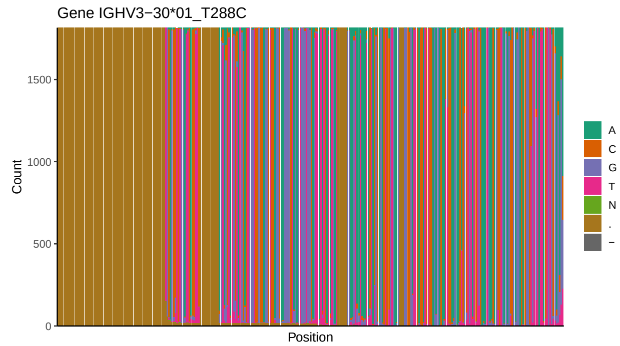

Gene IGHV4−61*01_i04_G88A_A106G_G129C_C134A_A147C_T165C
% share
100%
75%
50%
25%
0%
Contributing Reads
2500
2000
1500
1000
500
0
G C G A G A G A
315
320
Position
A
C
G
T

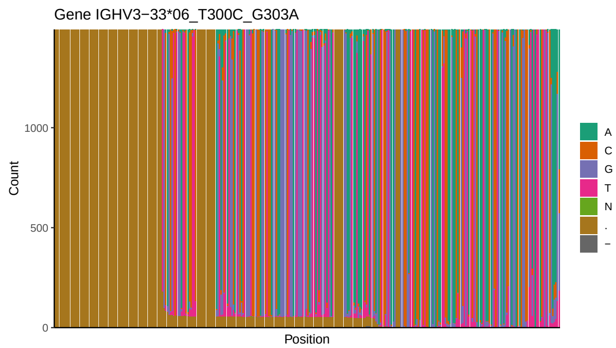

Gene IGHV4−61*02_A291G
% share
100%
75%
50%
25%
0%
Contributing Reads
3000
2000
1000
0
G C G A G A G A
315
320
Position
A
C
G
T

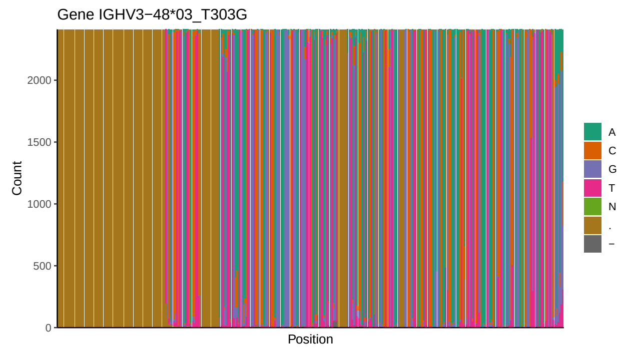

Gene IGHV4−NL1*01_C300T
% share
100%
75%
50%
25%
0%
Contributing Reads
2000
1500
1000
500
0
G C G A G A C A
315
320
Position
A
C
G
T

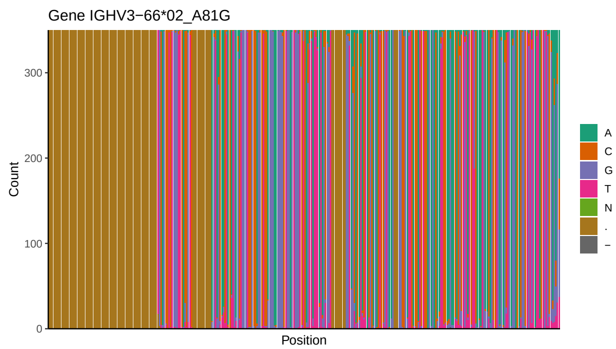
1.3 Whole-sequence composition of each assigned read

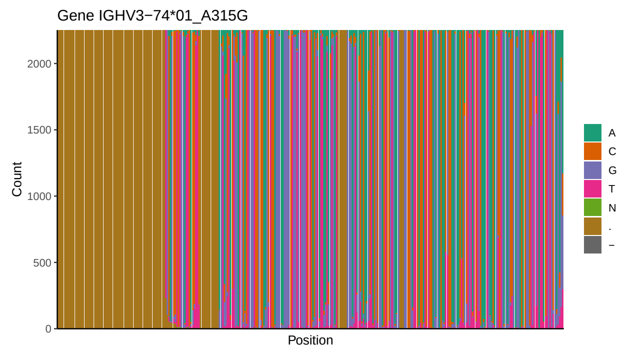


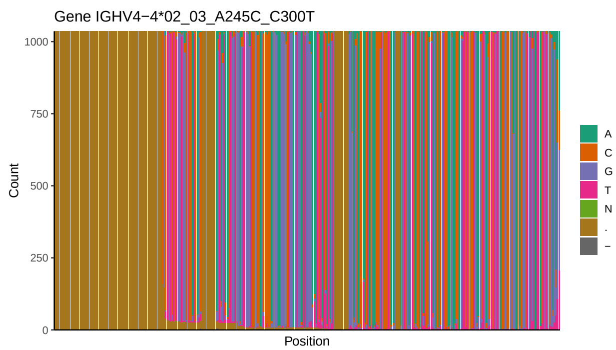


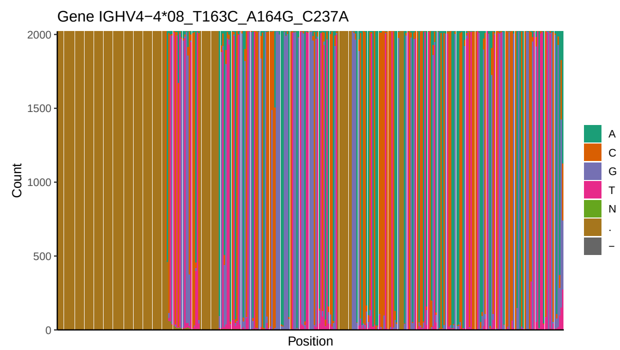


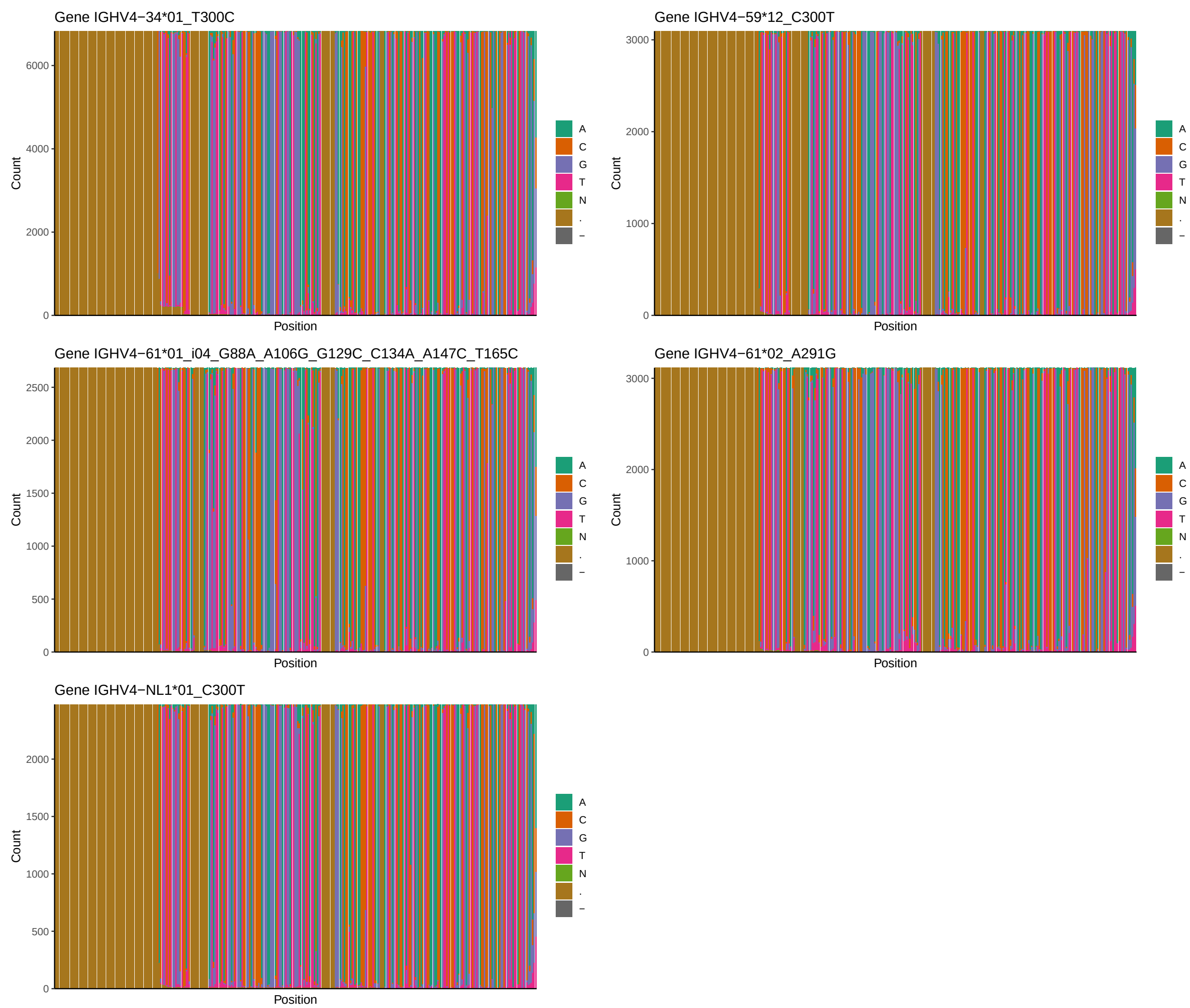




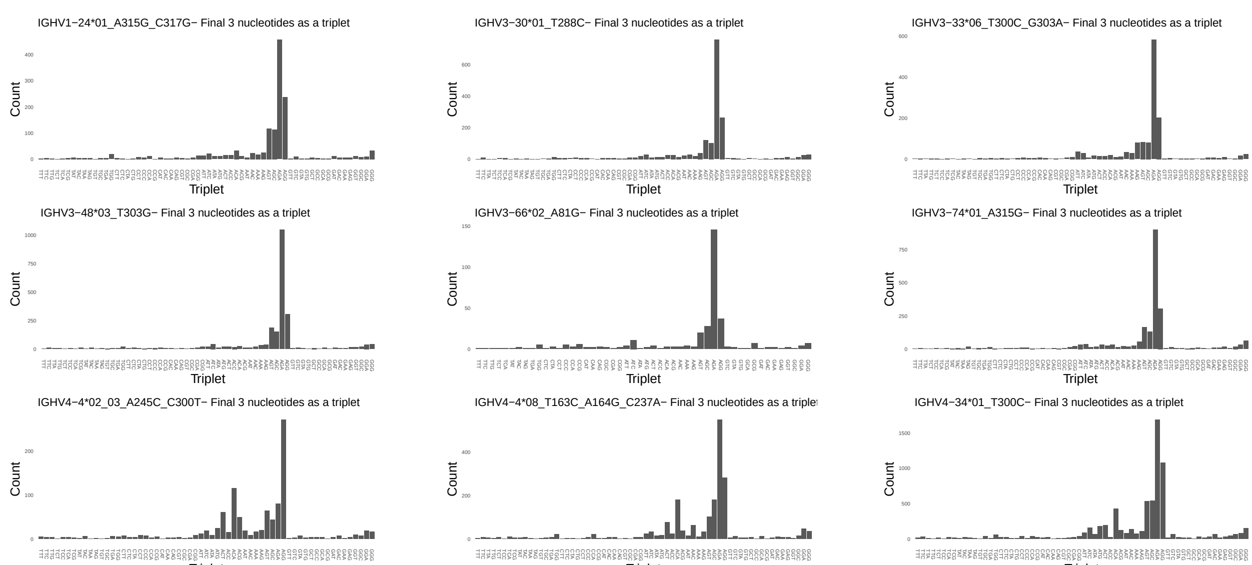


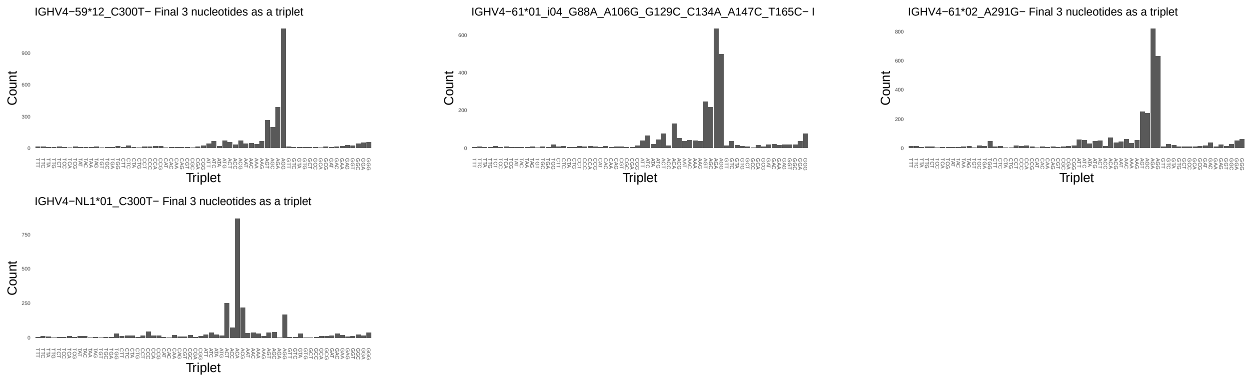




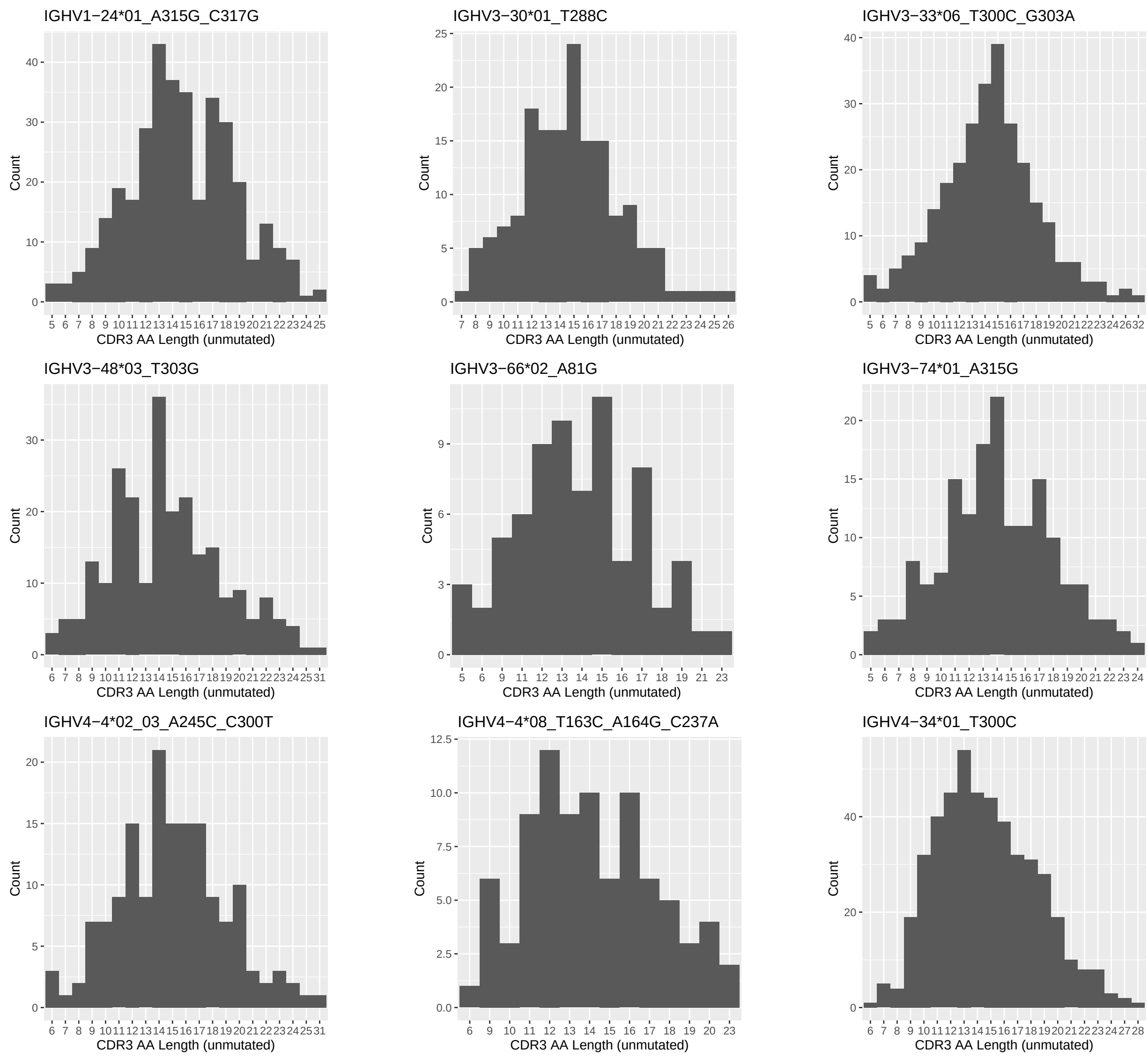


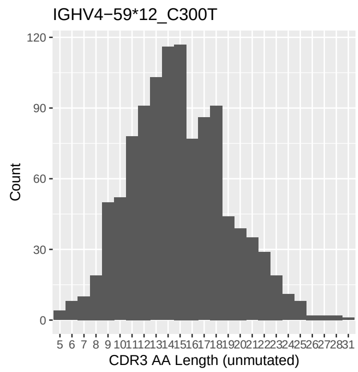
1.4 Final three nucleotides: frequency of each observed triplet

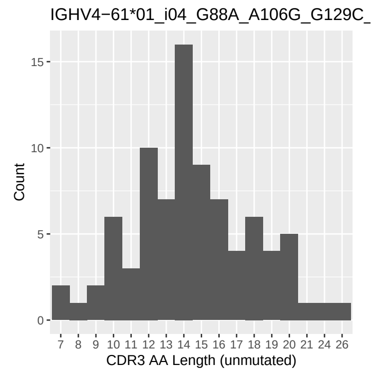


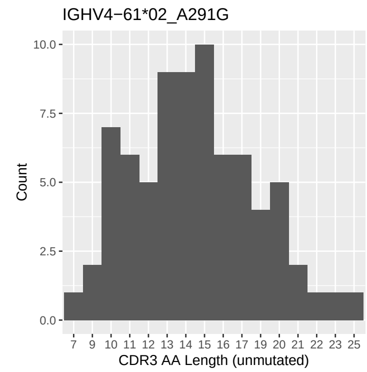


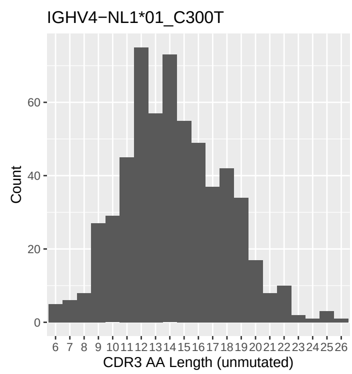
1.5 CDR3 length distribution, in assignments to novel alleles



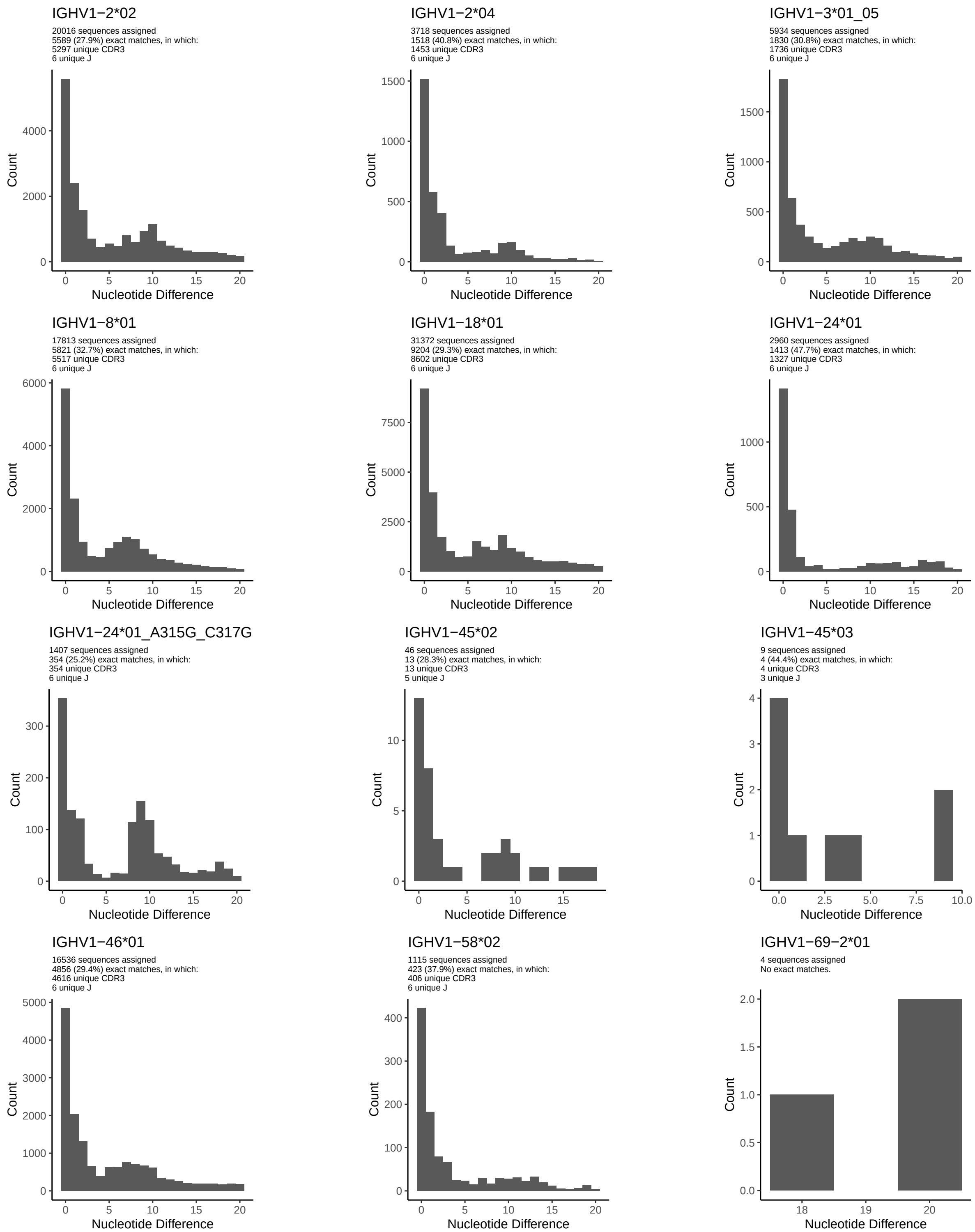

IGHV4−59*12_C300T
Count
120
90
60
30
0
5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 31
CDR3 AA Length (unmutated)

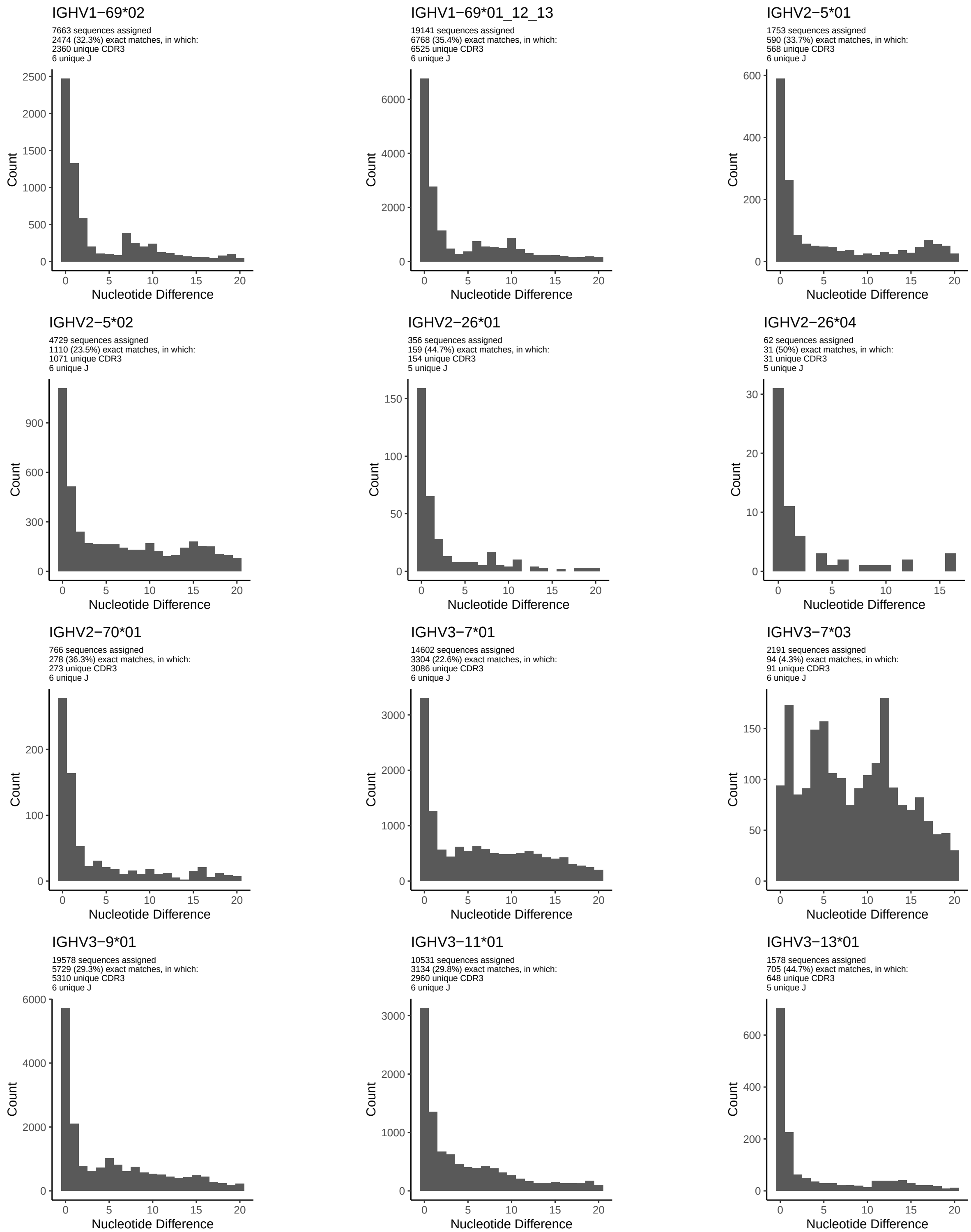

IGHV4−61*01_i04_G88A_A106G_G129C_
Count
15
10
5
0
7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 24 26
CDR3 AA Length (unmutated)

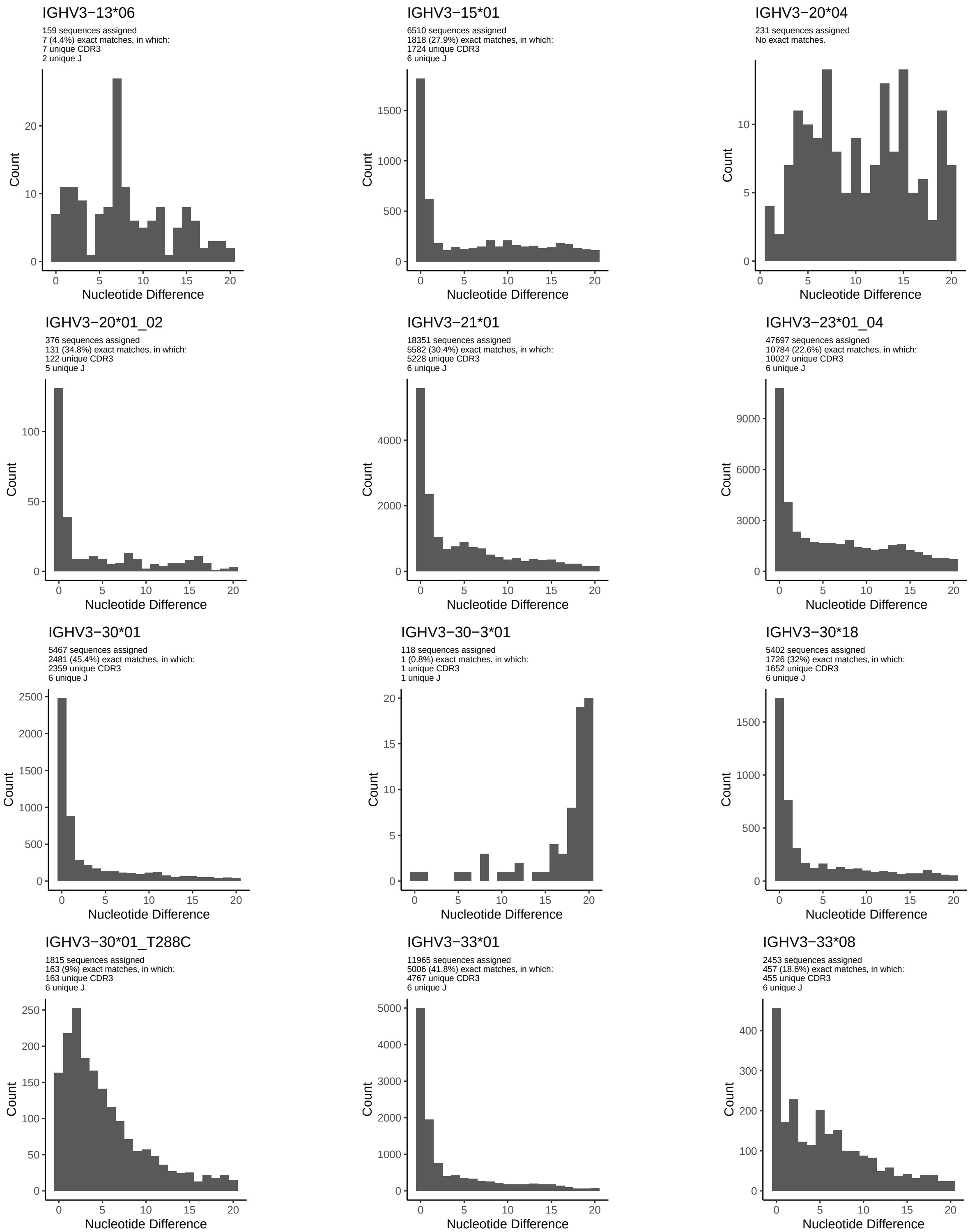

IGHV4−61*02_A291G
Count
10.0
7.5
5.0
2.5
0.0
7 9 10 11 12 13 14 15 16 17 19 20 21 22 23 25
CDR3 AA Length (unmutated)

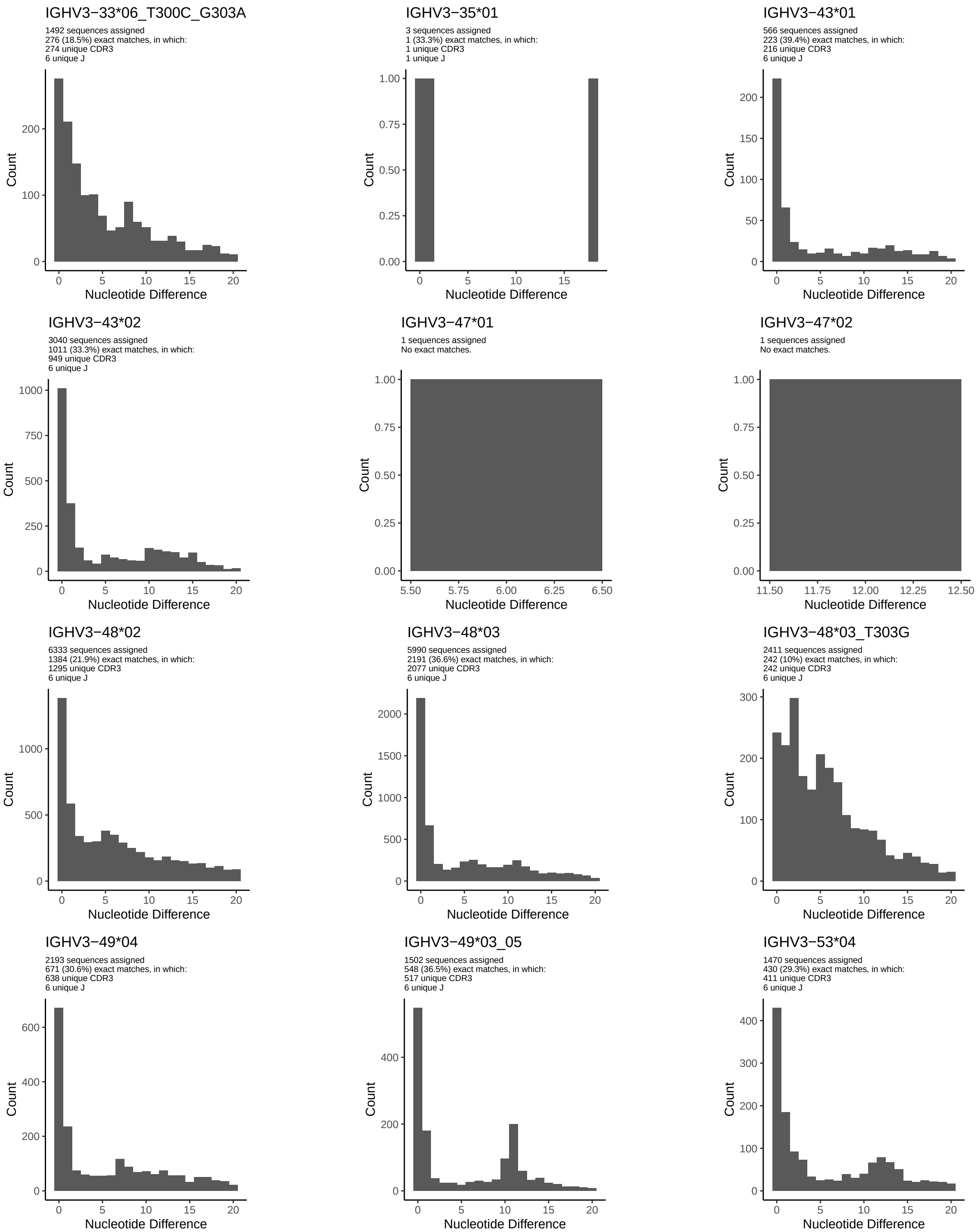

IGHV4−NL1*01_C300T
Count
60
40
20
0
6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26
CDR3 AA Length (unmutated)

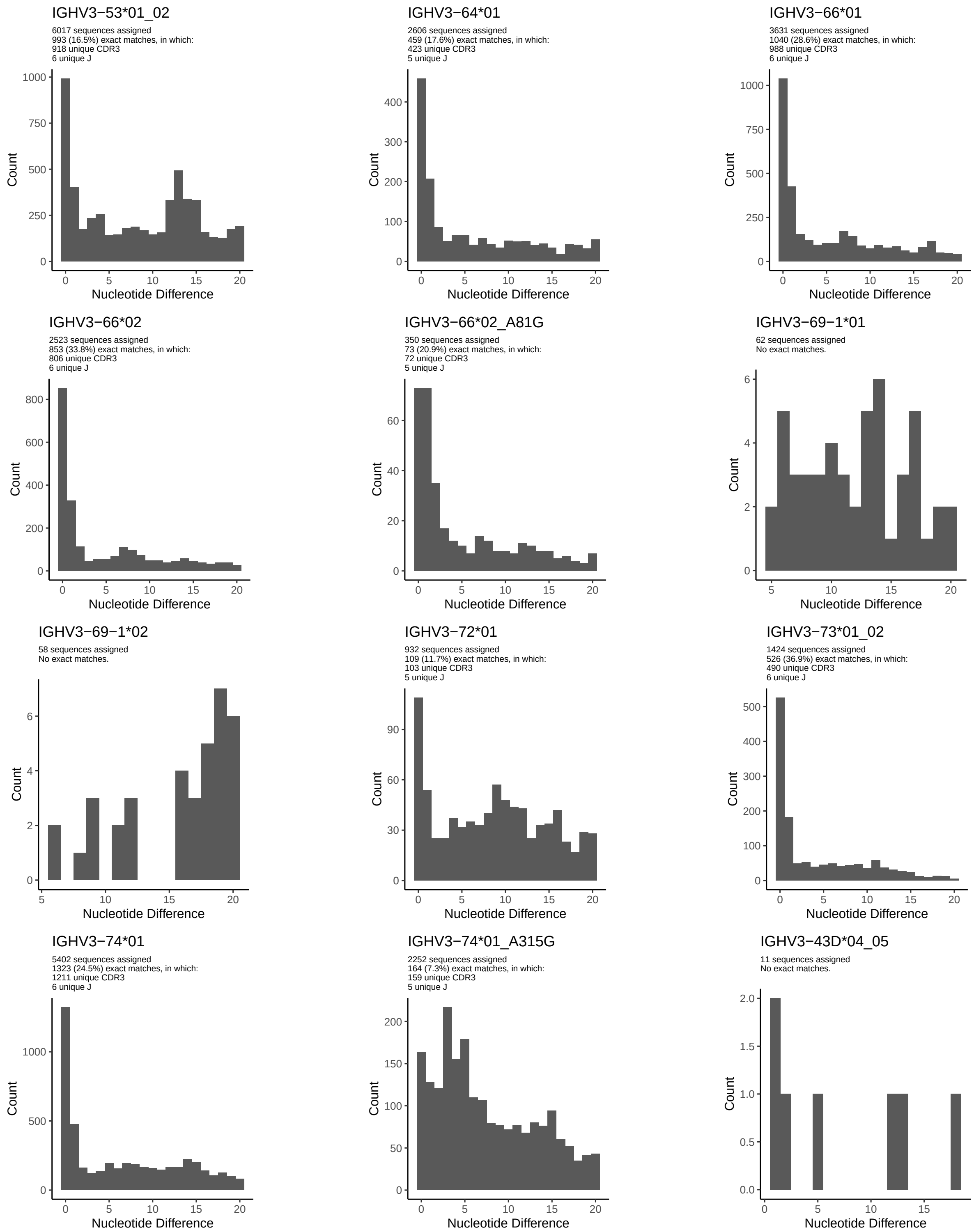
2 Variation from germline, in assignments to each allele



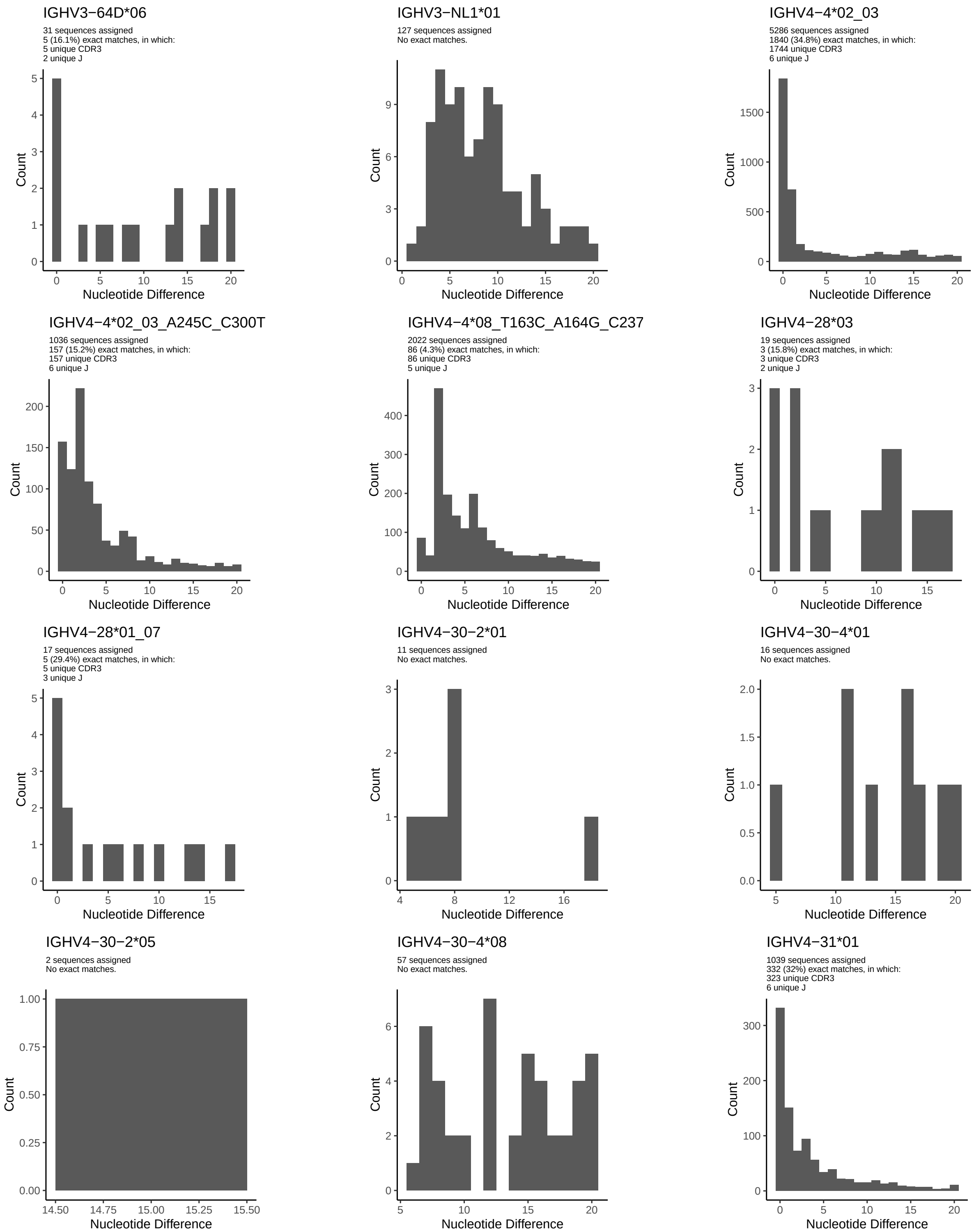

IGHV1−69*02
7663 sequences assigned
2474 (32.3%) exact matches, in which:
2360 unique CDR3
6 unique J
IGHV1−69*01_12_13
19141 sequences assigned
6768 (35.4%) exact matches, in which:
6525 unique CDR3
6 unique J
IGHV2−5*01
1753 sequences assigned
590 (33.7%) exact matches, in which:
568 unique CDR3
6 unique J
IGHV2−5*02
4729 sequences assigned
1110 (23.5%) exact matches, in which:
1071 unique CDR3
6 unique J
IGHV2−26*01
356 sequences assigned
159 (44.7%) exact matches, in which:
154 unique CDR3
5 unique J
IGHV2−26*04
62 sequences assigned
31 (50%) exact matches, in which:
31 unique CDR3
5 unique J
IGHV2−70*01
766 sequences assigned
278 (36.3%) exact matches, in which:
273 unique CDR3
6 unique J
IGHV3−7*01
14602 sequences assigned
3304 (22.6%) exact matches, in which:
3086 unique CDR3
6 unique J
IGHV3−7*03
2191 sequences assigned
94 (4.3%) exact matches, in which:
91 unique CDR3
6 unique J
IGHV3−9*01
19578 sequences assigned
5729 (29.3%) exact matches, in which:
5310 unique CDR3
6 unique J
IGHV3−11*01
10531 sequences assigned
3134 (29.8%) exact matches, in which:
2960 unique CDR3
6 unique J
IGHV3−13*01
1578 sequences assigned
705 (44.7%) exact matches, in which:
648 unique CDR3
5 unique J
Count
Nucleotide Difference

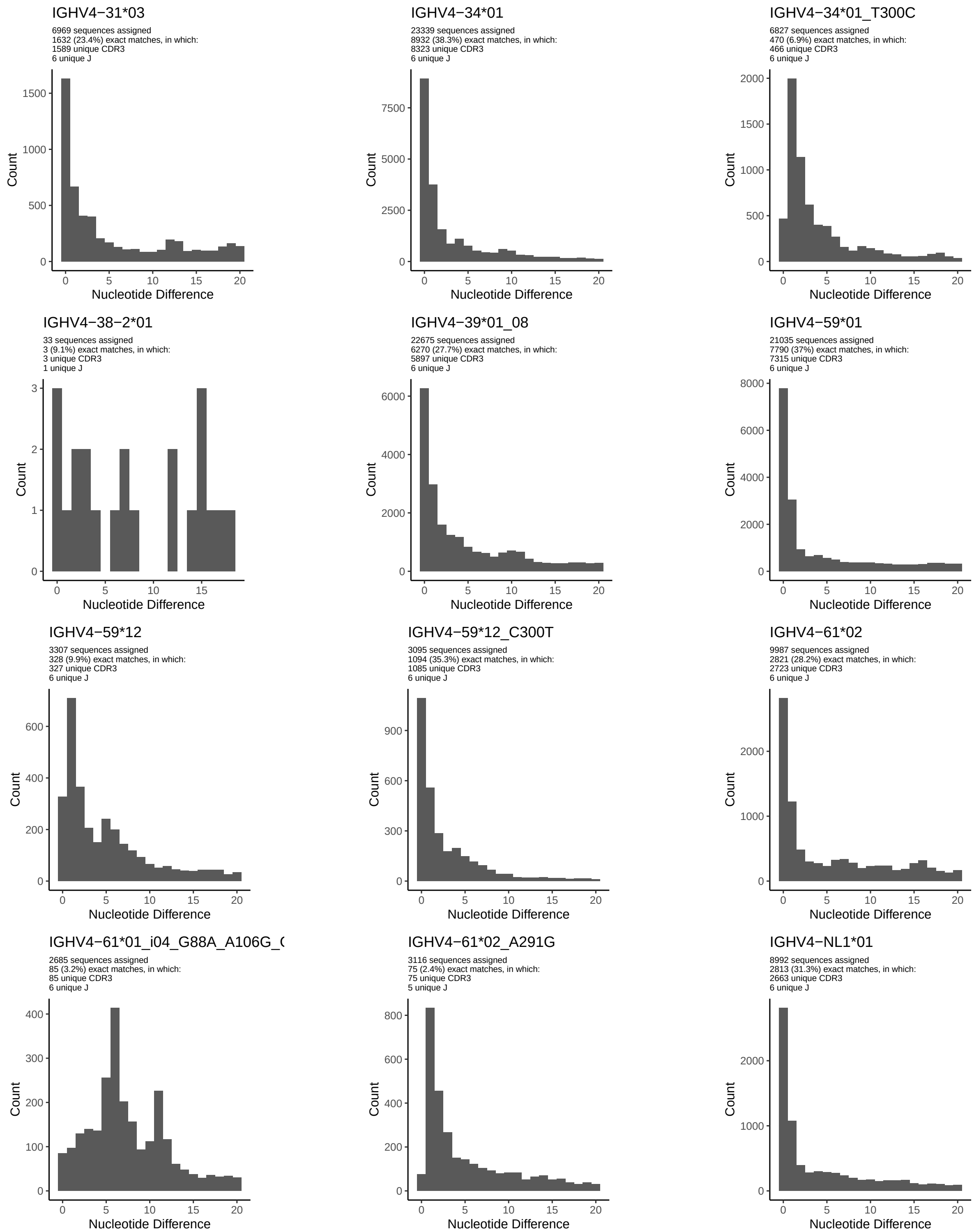

IGHV3−13*06
159 sequences assigned
7 (4.4%) exact matches, in which:
7 unique CDR3
2 unique J
IGHV3−15*01
6510 sequences assigned
1818 (27.9%) exact matches, in which:
1724 unique CDR3
6 unique J
IGHV3−20*04
231 sequences assigned
No exact matches.
IGHV3−20*01_02
376 sequences assigned
131 (34.8%) exact matches, in which:
122 unique CDR3
5 unique J
IGHV3−21*01
18351 sequences assigned
5582 (30.4%) exact matches, in which:
5228 unique CDR3
6 unique J
IGHV3−23*01_04
47697 sequences assigned
10784 (22.6%) exact matches, in which:
10027 unique CDR3
6 unique J
IGHV3−30*01
5467 sequences assigned
2481 (45.4%) exact matches, in which:
2359 unique CDR3
6 unique J
IGHV3−30−3*01
118 sequences assigned
1 (0.8%) exact matches, in which:
1 unique CDR3
1 unique J
IGHV3−30*18
5402 sequences assigned
1726 (32%) exact matches, in which:
1652 unique CDR3
6 unique J
IGHV3−30*01_T288C
1815 sequences assigned
163 (9%) exact matches, in which:
163 unique CDR3
6 unique J
IGHV3−33*01
11965 sequences assigned
5006 (41.8%) exact matches, in which:
4767 unique CDR3
6 unique J
IGHV3−33*08
2453 sequences assigned
457 (18.6%) exact matches, in which:
455 unique CDR3
6 unique J
Count
Nucleotide Difference

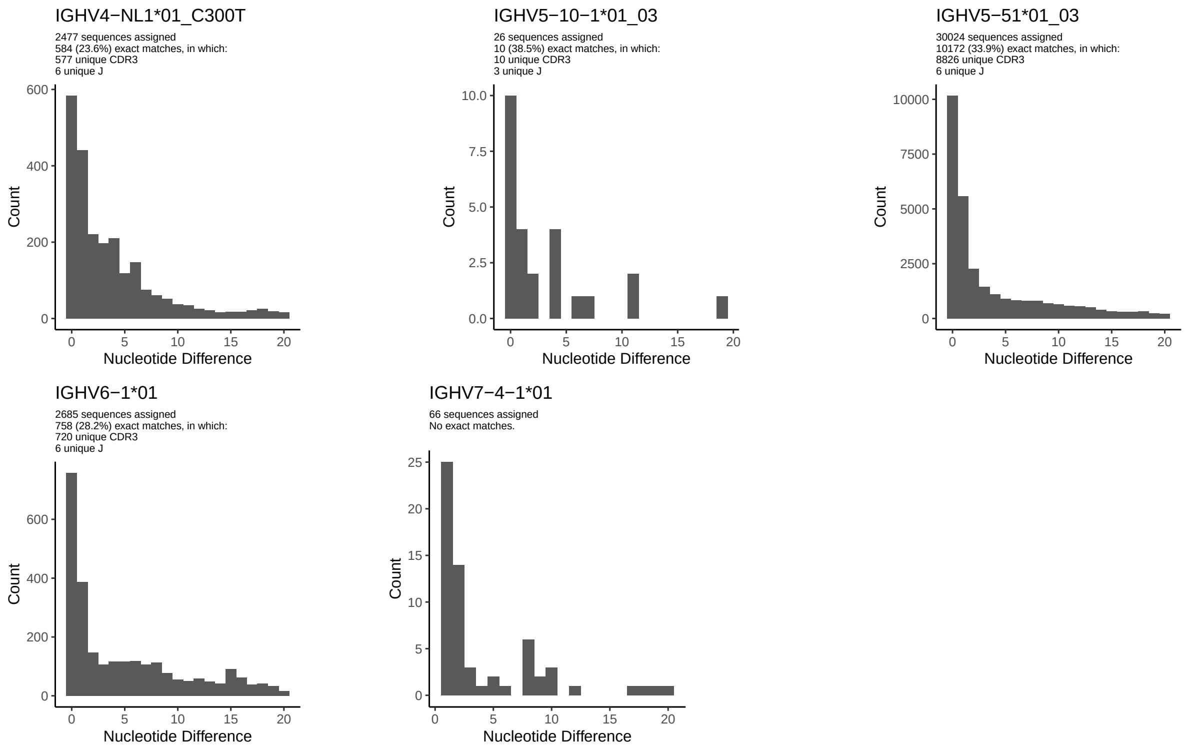

IGHV3−33*06_T300C_G303A
1492 sequences assigned
276 (18.5%) exact matches, in which:
274 unique CDR3
6 unique J
IGHV3−35*01
3 sequences assigned
1 (33.3%) exact matches, in which:
1 unique CDR3
1 unique J
IGHV3−43*01
566 sequences assigned
223 (39.4%) exact matches, in which:
216 unique CDR3
6 unique J
IGHV3−43*02
3040 sequences assigned
1011 (33.3%) exact matches, in which:
949 unique CDR3
6 unique J
IGHV3−47*01
1 sequences assigned
No exact matches.
IGHV3−47*02
1 sequences assigned
No exact matches.
IGHV3−48*02
6333 sequences assigned
1384 (21.9%) exact matches, in which:
1295 unique CDR3
6 unique J
IGHV3−48*03
5990 sequences assigned
2191 (36.6%) exact matches, in which:
2077 unique CDR3
6 unique J
IGHV3−48*03_T303G
2411 sequences assigned
242 (10%) exact matches, in which:
242 unique CDR3
6 unique J
IGHV3−49*04
2193 sequences assigned
671 (30.6%) exact matches, in which:
638 unique CDR3
6 unique J
IGHV3−49*03_05
1502 sequences assigned
548 (36.5%) exact matches, in which:
517 unique CDR3
6 unique J
IGHV3−53*04
1470 sequences assigned
430 (29.3%) exact matches, in which:
411 unique CDR3
6 unique J
Count
Nucleotide Difference



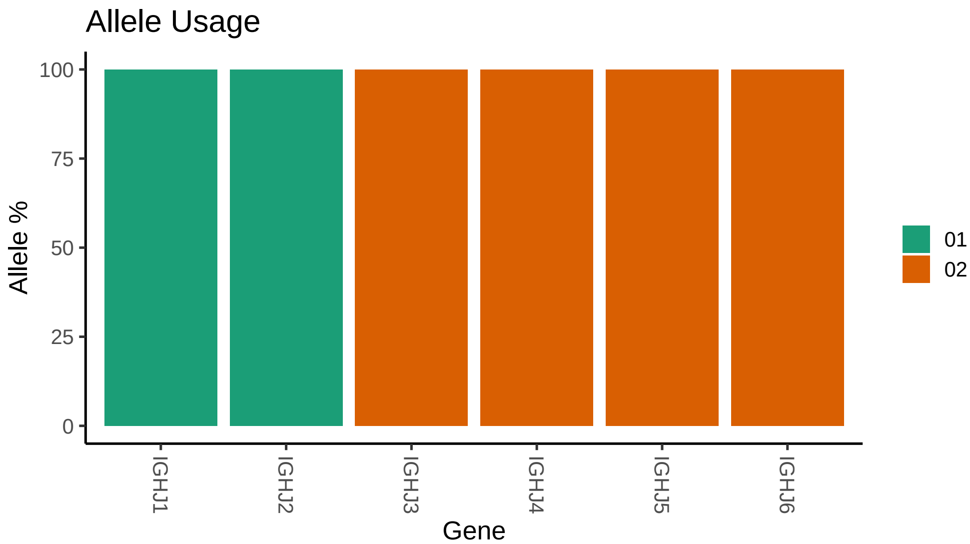
IGHV3−53*01_02
6017 sequences assigned
993 (16.5%) exact matches, in which:
918 unique CDR3
6 unique J
IGHV3−64*01
2606 sequences assigned
459 (17.6%) exact matches, in which:
423 unique CDR3
5 unique J
IGHV3−66*01
3631 sequences assigned
1040 (28.6%) exact matches, in which:
988 unique CDR3
6 unique J
IGHV3−66*02
2523 sequences assigned
853 (33.8%) exact matches, in which:
806 unique CDR3
6 unique J
IGHV3−66*02_A81G
350 sequences assigned
73 (20.9%) exact matches, in which:
72 unique CDR3
5 unique J
IGHV3−69−1*01
62 sequences assigned
No exact matches.
IGHV3−69−1*02
58 sequences assigned
No exact matches.
IGHV3−72*01
932 sequences assigned
109 (11.7%) exact matches, in which:
103 unique CDR3
5 unique J
IGHV3−73*01_02
1424 sequences assigned
526 (36.9%) exact matches, in which:
490 unique CDR3
6 unique J
IGHV3−74*01
5402 sequences assigned
1323 (24.5%) exact matches, in which:
1211 unique CDR3
6 unique J
IGHV3−74*01_A315G
2252 sequences assigned
164 (7.3%) exact matches, in which:
159 unique CDR3
5 unique J
IGHV3−43D*04_05
11 sequences assigned
No exact matches.
Count
Nucleotide Difference


IGHV3−64D*06
31 sequences assigned
5 (16.1%) exact matches, in which:
5 unique CDR3
2 unique J
IGHV3−NL1*01
127 sequences assigned
No exact matches.
IGHV4−4*02_03
5286 sequences assigned
1840 (34.8%) exact matches, in which:
1744 unique CDR3
6 unique J
IGHV4−4*02_03_A245C_C300T
1036 sequences assigned
157 (15.2%) exact matches, in which:
157 unique CDR3
6 unique J
IGHV4−4*08_T163C_A164G_C237
2022 sequences assigned
86 (4.3%) exact matches, in which:
86 unique CDR3
5 unique J
IGHV4−28*03
19 sequences assigned
3 (15.8%) exact matches, in which:
3 unique CDR3
2 unique J
IGHV4−28*01_07
17 sequences assigned
5 (29.4%) exact matches, in which:
5 unique CDR3
3 unique J
IGHV4−30−2*01
11 sequences assigned
No exact matches.
IGHV4−30−4*01
16 sequences assigned
No exact matches.
IGHV4−30−2*05
2 sequences assigned
No exact matches.
IGHV4−30−4*08
57 sequences assigned
No exact matches.
IGHV4−31*01
1039 sequences assigned
332 (32%) exact matches, in which:
323 unique CDR3
6 unique J
Count
Nucleotide Difference


IGHV4−31*03
6969 sequences assigned
1632 (23.4%) exact matches, in which:
1589 unique CDR3
6 unique J
IGHV4−34*01
23339 sequences assigned
8932 (38.3%) exact matches, in which:
8323 unique CDR3
6 unique J
IGHV4−34*01_T300C
6827 sequences assigned
470 (6.9%) exact matches, in which:
466 unique CDR3
6 unique J
IGHV4−38−2*01
33 sequences assigned
3 (9.1%) exact matches, in which:
3 unique CDR3
1 unique J
IGHV4−39*01_08
22675 sequences assigned
6270 (27.7%) exact matches, in which:
5897 unique CDR3
6 unique J
IGHV4−59*01
21035 sequences assigned
7790 (37%) exact matches, in which:
7315 unique CDR3
6 unique J
IGHV4−59*12
3307 sequences assigned
328 (9.9%) exact matches, in which:
327 unique CDR3
6 unique J
IGHV4−59*12_C300T
3095 sequences assigned
1094 (35.3%) exact matches, in which:
1085 unique CDR3
6 unique J
IGHV4−61*02
9987 sequences assigned
2821 (28.2%) exact matches, in which:
2723 unique CDR3
6 unique J
IGHV4−61*01_i04_G88A_A106G_(
2685 sequences assigned
85 (3.2%) exact matches, in which:
85 unique CDR3
6 unique J
IGHV4−61*02_A291G
3116 sequences assigned
75 (2.4%) exact matches, in which:
75 unique CDR3
5 unique J
IGHV4−NL1*01
8992 sequences assigned
2813 (31.3%) exact matches, in which:
2663 unique CDR3
6 unique J
Count
Nucleotide Difference


IGHV4−NL1*01_C300T
2477 sequences assigned
584 (23.6%) exact matches, in which:
577 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV5−10−1*01_03
26 sequences assigned
10 (38.5%) exact matches, in which:
10 unique CDR3
3 unique J
Count
Nucleotide Difference
IGHV5−51*01_03
30024 sequences assigned
10172 (33.9%) exact matches, in which:
8826 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV6−1*01
2685 sequences assigned
758 (28.2%) exact matches, in which:
720 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV7−4−1*01
66 sequences assigned
No exact matches.
Count
Nucleotide Difference

3 Allele usage in potential haplotype anchor genes



4 Haplotype plots

5 Configuration settings

```
## Repertoire file: /work/jenkins/PRJNA491287/M64 113/M64 113/M64 113/M64 113/b8/a51ac23e50a166bd5ee3a6
##
## Germline reference file: /work/jenkins/PRJNA491287/M64 113/M64 113/M64 113/M64 113/b8/a51ac23e50a166b
##
## Novel allele file: /work/jenkins/PRJNA491287/M64 113/M64 113/M64 113/M64 113/b8/a51ac23e50a166bd5ee3
##
## Species: Homosapiens
##
## Chain: IGHV
##
## Segment: V
##
## Unexpected N in reference sequence IGHV1 24 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 24 01_A315G_C317G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 30 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 01_T288C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 33 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 33 06_T300C_G303A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 48 03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 48 03_T303G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 66 02: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 66 02_A81G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 74 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 74 01_A315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 4 02_03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 4 02_03_A245C_C300T: replacing with gap
```

```
##
##
## Unexpected N in reference sequence IGHV4 61 02: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 4 08_T163C_A164G_C237A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 34 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 34 01_T300C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 59 12: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 59 12_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 31 03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 61 01_i04_G88A_A106G_G129C_C134A_A147C_T165C: replacing with ga
##
##
## Unexpected N in novel sequence IGHV4 61 02_A291G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 NL1 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 NL1 01_C300T: replacing with gap
```